MIL-D-2477C

29 February 1968
SUPERSEDING
MIL-D-2477B
12 March 1963

MILITARY SPECIFICATION

DOUGH MOLDING MACHINES, LOAF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers two types of electrically operated dough molding machines for molding dough into predetermined uniform size loaves.

1.2 Classification.- Dough molding machine, loaf shall be of the following types as specified (see 6.2).

Type I - Dough molding machine with cross-grain molding.
Type II - Dough molding machine without cross-grain molding.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-636 - Motor, Alternating Current, (Fractional Horsepower)
CC-M-641 - Motor, Alternating Current, (Integral Horsepower, 200 HP and Smaller)

MILITARY

MIL-B-2316 - Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery Of

FSC 7320

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of

(Copies of specifications, standards and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publication.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

BAKING INDUSTRY SANITATION STANDARD (BISSC)

Sanitation Standard No. 9 for Bread Molders 1967

(Copies of Baking Industry Sanitation Standards Publication may be obtained from the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, N.Y. 10017.)

UNDERWRITER'S LABORATORIES, INC. (UL)

Electrical Construction Materials List

(Copies may be obtained from Underwriters; Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611, Walt Whitman Road, Melville, L.I., N.Y. 11749, and 1655 Scott Boulevard, Santa Clara, California 95050.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced, a sample dough molding machine covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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3.2 Standard product.- The dough molding machine delivered under this specification shall be of the manufacturer's commercially current standard product except for any change necessary to comply with specification requirements.

* 3.3 Standards.- All electrical equipment shall conform to UL Standards as applicable and the complete dough molding machine shall conform to the requirements of BISSC Standard No. 9 for Bread Moulders.

* 3.3.1 Certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the electrical components and the dough molding machine(s) he proposes to furnish under this specification meet the requirements of UL Standards as applicable and BISSC Standard No. 9 respectively.

* 3.3.2 UL.- Acceptable evidence of meeting the applicable requirements of UL Standards shall be a UL label; a listing in the "Electrical Appliance and Utilization Equipment List", or a certified test report from a recognized independent testing laboratory, acceptable to the Government indicating that the electric components conform to the applicable UL Standards.

3.3.3 BISSC.- Acceptable evidence of meeting the applicable requirements of BISSC Standard No. 9 shall be the BISSC Seal of approval on the dough molder or a certified test report from a recognized independent testing laboratory, acceptable to the medical department of the procuring service indicating that the dough molders offered conform to the applicable requirements of BISSC Standard No. 9.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used in the manufacture of items of this type.

3.4.1 Motors.- Electric motors shall conform to CC-M-636 or CC-M-641, as applicable. Motors shall be of the continous-duty, induction type, totally inclosed, without commutation devices, having not more than one starting contact.

3.5 Design.- The molding machines shall be designed to receive pieces of dough from an intermediate proofer and shape the dough into loaves ready to be placed into pans. The machines shall be capable of running at variable speeds. The type I and type II machines shall mold 22-ounce dough pieces at a rate of 60 pieces a minute and 36-ounce dough pieces at a rate of 40 pieces a minute. The type I molder shall be equipped with two conveyor belts perpendicular to each other, which shall permit the dough to be sheeted and then transferred to the other conveyor which shall permit the dough to be curled and molded for panning. The type II molder shall be capable of producing uniform dough pieces which have been sheeted and molded.

3.6 Construction.- The machine frame shall be constructed with cast iron or steel. Shafts shall be made of steel, and shall prevent lateral and torsional deflections and maintain the close clearances between mating parts of the rolls.

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All bearings shall be of a replaceable type, and shall be protected against flour dust and sealed to prevent loss of lubricant. When tested as specified in 4.5.1, moving parts shall have no undue vibration, bearings shall not overheat and assembly of the component parts shall be complete, of proper interrelationship, and the starting and stopping and adjusting devices shall be positive in action and operate as intended.

3.6.1 Sheeting head.- The sheeting head shall be of the integral type. The rolls and molding belt speeds shall be correlated to effectively handle the production requirements specified in 3.5.

3.6.2 Number of pairs of rolls.- The machine shall be provided with three pairs of rolls or be of a type with two pairs of rolls in the head proper and a third set of rolls superimposed over the head by means of an auxiliary pre-sheeting device.

3.6.3 Rolls.- The rolls shall be made of hardened and ground alloy steel, or polytetrafluoroethylene or trifluorochloroethylene, and shall be mounted on shafts with provision for replacement. Front rolls shall be plain, and the rear rolls shall have side flanges of sufficient height to handle the full weight range of dough pieces. Rolls shall be of the correct width to sheet dough pieces for molded lengths as specified in 3.6.8. Rolls shall have independent adjustment for sheet thickness or two pairs may be controlled by common adjustment. Provisions shall be made to accurately indicate in thousandths of an inch the openings between the mating pairs of rolls.

3.6.4 Roll scrapers.- Roll scrapers shall be made of hardened and ground materials and may be coated with tetrafluoroethylene or trifluorochloroethylene. They shall be mounted on adjustable supports to provide proper scraping action without heating, excessive wear, vibration, or noise.

3.6.5 Dusters.- Each machine shall be provided with two dusters, complete with adjustments for flour delivery and distribution, and with suitable dust guides or baffles to direct the flour to the head rolls and on the molding belt.

3.6.6 Curling chain.- The curling chain shall be made of corrosion-resisting steel wire mesh designed to prevent slippage of sheeted dough piece and shall be adjustable.

3.6.7 Pressure plate.- The pressure plate shall be independently adjustable at both ends. The plate shall be adjustable to a minimum of 1-1/2 inches above the moulding belt.

3.6.8 Side guides.- Each pressure chamber shall be equipped with two side guides. Guides shall be adjustable vertically to restrict pilling and horizontally to provide a finished loaf range of 7 to 17 inches in length.

3.6.9 Conveyor.- The molding conveyor belt shall be made of solid woven cotton, 3- or 4-ply construction. Means shall be provided for allowing a minimum adjustment of 1-1/2 inches between the centerline of the molding-belt rollers to allow for adjusting the belt tension.

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3.6.10 Take-off apron.- A take-off apron of polished corrosion-resisting steel shall be provided at the discharge end of the molding belt to receive the molded dough pieces.

3.6.11 Electrical requirements.-

3.6.11.1 Electric motors.- The motors shall have ample power to operate the machine and components continously at maximum capacity without exceeding the allowable temperature rise. Motors shall be as specified in 3.4.1, as applicable. The molder shall be completely wired and equipped with a conveniently mounted push-button station and magnetic switch with thermal overload protection. The pushbutton station shall be located, so that it is accessible to an operator standing at the take off position.

3.6.11.2 Electrical rating.- Unless otherwise specified (see 6.2), all electrical equipment shall be designed for operation on a nominal 208-volt, alternating current (AC), 60-cycle, three-phase power system.

* 3.7 Radio interference suppression.- Equipment processed under this specification shall have radio interference suppression conforming to class IIB of MIL-STD-461 (see 4.5.2).

3.8 Lubrication.- Bearings and all other parts requiring pressure lubrication shall be provided with pressure - type grease fittings. Close oil cups shall be provided on all fittings which require oil lubrication. At time of delivery, the unit shall be completely lubricated with the manufacturer's standard lubricant.

3.8.1 Lubrication chart.- The supplier shall affix to each machine a corrosion-resistant metal lubricating chart indicating:

(a) All points of the equipment requiring lubrication.
(b) Grade and type of lubricant required.
(c) Time interval when lubrication should be accomplished.

3.8.2 Finish.- All exterior surfaces of the machines not coming in contact with dough, except corrosion-resisting steel shall be given a white enamel finish in accordance with the manufacturer's commercial practice.

3.9 Marking for identification.- The dough molding machine shall have the letters "U.S." and additional pertinent rating characteristics or information necessary for the safe handling operation and maintenance of the items on a corrosion resistant metal plate in accordance with MIL-STD-130. The name plate shall be visible to the eye and shall be securely affixed in a conspicuous location.

3.10 Equipment manual.- The supplier shall furnish with each dough molder with an equipment manual. Manual contents and additional distribution shall be as specified in the contract or order (see 6.2).

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3.11 Repair and maintenance.- Provisions shall be made to allow for ready adjustment, service and replacement of components without disturbing or dissassembly of any major components. In addition there shall be froo acess to inspect, service, and adjust any component or equipment as provided for by this specification or as indicated by the component or equipment manufacturer, without requiring disassembly of any other component or equipment on the dough molder.

3.12 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.13 Workmanship.-

* 3.13.1 Fabrication.- Metal used shall be free from fractures, splits, punctures, dents, deterioration and shall not be bowed, or malformed. There shall be no sharp burr, sliver or splinter.

* 3.13.2 Welding and brazing.- All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures, undercuts, or incomplete fusion. All scale or flux deposits shall be removed from finished welds. Welds shall be smooth and uniform.

* 3.13.3 Solder.- All surfaces to be soldered (when applicable) shall be thoroughly cleaned prior to soldering. Solder joints shall be smooth with no pin holes.

* 3.13.4 Threaded fasteners.- Threaded fasteners shall not be loose or missing. Threads shall not be missing, stripped or broken.

* 3.13.5 Electrical assembly.- Wiring shall (1) be properly joined, (2) not be cut or abraded, (3) not be loose at terminal end, (4) be properly inclosed in conduit when required, (5) be provided with slack, and (6) not be stripped of required insulation.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproductions sample is required, it shall be examined for defects listed in table I and tested as specified in 4.5.1 and 4.5.2.

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4.3.1 Component and material inspection.- In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none in accordance with this specifiaction.

4.3.2 End item inspection.- A lot shall consist of all dough molding machines of the same type and model offered for inspection at one time. The sample unit shall be one completely fabricated dough molding machine.

* 4.3.2.1 Visual examination.- The end item shall be examined for the defects set forth in table I. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Type not as specified	X	
	Blistered, peeled chipped or area of no film or thin film		X
	Not color specified		X
	Stained, overspray, wet or tacky paint, foreign matter imbedded in finish, or evidence of rust	X	
Construction and workmanship, general (applicable to all components and assemblies)	Surfaces not smooth machined finish (where specified)	X	
	Part fractured, split, torn, dented, bowed, sprung, miscast, or otherwise impaired	X	
	Part misplaced, loose or not proper alignment	X	
	Any operation omitted or not properly performed	X	
	Sharp burr, sliver, or edge that may cause injury	X	
	Functioning component that requires abnormal effort to operate	X	
	Functioning component that is inoperative or will not operate as intended	X	
	Bearings and moving parts not protectected from dust and oil	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship, general (applicable to all components and assemblies)	Moving parts not inclosed or equipped with safety guards	X	
	Rollers not easily adjustable	X	
	Safety guard to protect hands not provided	X	
	Switch not positive in action and conveniently located		X
	Machine not bolted to stand	X	
	Machine not clean		X
Welding and brazing	Surface to be welded not free from oxide, scale, paint, grease, or other foreign matter		X
	Not continous, not sound, not smooth, and not free from porosity, cracks, or incomplete fusion	X	
	Scale or flux deposit not removed		X
Castings (when specified)	Blowhole, porosity, or crack	X	
Bolts, nuts, and studs	Missing, not approved type, broken, stripped, fractured, or loose		X
Electrical apparatus and wiring	Wiring cut, abraded, loose at terminal, not properly joined, or otherwise impaired	X	
Motors	Not type required	X	
Lubrication	Not lubricated where required		X
	Insufficiently lubricated		X
	Fitting missing or not specified type	X	
	Fitting defective	X	
Dimensions	Not as specified		X
Identification Marking	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in manner specified or not correct		X

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4.3.3 Testing of the end item.- When a preproduction sample is not required, the first production unit shall be tested as specified in 4.5.2. Each dough molding machine shall be tested as specified in 4.5.1. Any nonconformance shall be cause for rejection.

4.3.4 Examination of preparation for delivery.- Examination shall be made for preservation, packaging, packing and marking in accordance with MIL-B-2316.

4.4 Standard compliance.- Proof of compliance with the requirements of 3.3 and 3.3.1 shall be available for the Government representative.

4.5 Test.-

4.5.1 Operational Test.- Each dough molding machine shall be connected to power and operated without load for a minimum of 15 minutes to determine that all operating components and systems are in operable condition to determine compliance with 3.6.

* 4.5.2 Electromagnetic compatibility tests.- The preproduction sample or the initial production unit shall be equipped for the reduction of radio interference in accordance with 3.7 and will be tested by the supplier or his private testing organization in accordance with Test Methods CE03, CE05, and RE02 of Military Standard 462. The supplier shall furnish the contracting officer, for approval, the Interference Control Plan, the Electromagnetic Compatibility Test Report and the EMI/EMC Test Plan required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met the preproduction sample or the initial production unit shall be used as a model for all production units.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, or C.- Preservation packaging, packing and marking shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use.- The dough molding machines covered by this specification are intended for use in garrison bakeries.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type machine required (see 1.2)
(c) When preproduction sample is not required (see 3.1)
(d) Electrical characteristic if other than specified (see 3.6.11.2)
(e) Manual contents and quantity required (see 3.10)
(f) Selection of the applicable levels of preservation and packaging and packing (see 5.1)
(g) Whether supervision of installation is to be provided (see 6.4)

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6.3 Electromagnetic compatibility.- The contracting officer should submit a copy of the report(s) required (see 4.5.2) to the Commanding General, U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703 for approval.

6.4 Supervision.- Supervision of installation should be required of the supplier of dough molding machine(s) shipped to any point in Continental U.S., except those dough molding machines designated for storage or warehouse stock (see 6.2). When supervision of installation is required, the supplier should furnish the services of a competent superintendent who shall be familiar with the construction, erection, iperation, maintenance and repair of the equipment and the required utility connections, Material required for complete installations other than that specified herein and all installation labor should be furnished by the Government. All utilities should be extended and connected to the equipment by the Government.

* 6.5 The margins of this specification are marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - EL, MD
Navy - YD

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 7320-0226

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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